Final report

RUN · F4B26D6A · SCORE MISSISSAUGA (L5B) FOR EV AND HEAT PUMP GROWTH THROUGH 2030

FSAS STUDIED	HOUSEHOLDS MODELED	SCENARIOS TESTED	AGENT RUNTIME 32S
1	19,147	3	

VERDICT (WORST CASE)

CONSTRAINED

3/3 pairings need intervention

INCREMENTAL PEAK LOAD

+157.5 MW

under 2035 net-zero (50% hp, 60% ev)

INTERVENTION COST

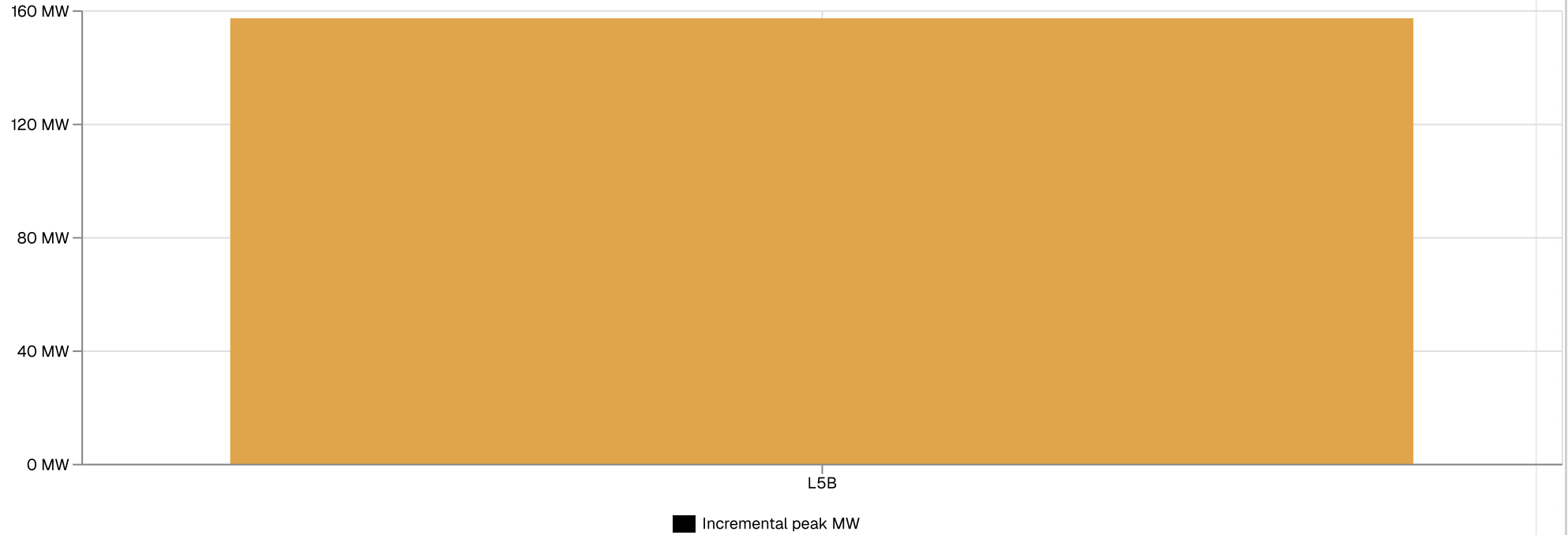
$9.0M

3 recommended action(s)

01 · PER-FSA LOAD IMPACT

Incremental winter-peak demand and panel-upgrade burden by neighborhood.

Worst-case scenario: 2035 Net-Zero (50% HP, 60% EV) · +157.5 MW total



Greater Toronto Area · L5B CONSTRAINED

INCREMENTAL PEAK

+157.5 MW (peak hour 20:00)

PER-HOUSEHOLD PEAK

10.0 kW

PANEL UPGRADES

0% of 19,147 homes

TRANSFORMER OVERLOADS

478

02 · FSA READINESS MAP

Neighborhoods colored by readiness verdict, sized by households modeled.

1 FSA plotted. Marker color = worst-case readiness, size = households modeled.



READY CONSTRAINED BLOCKED

Multi-dimensional readiness across all 3 (FSA × scenario) pairings.

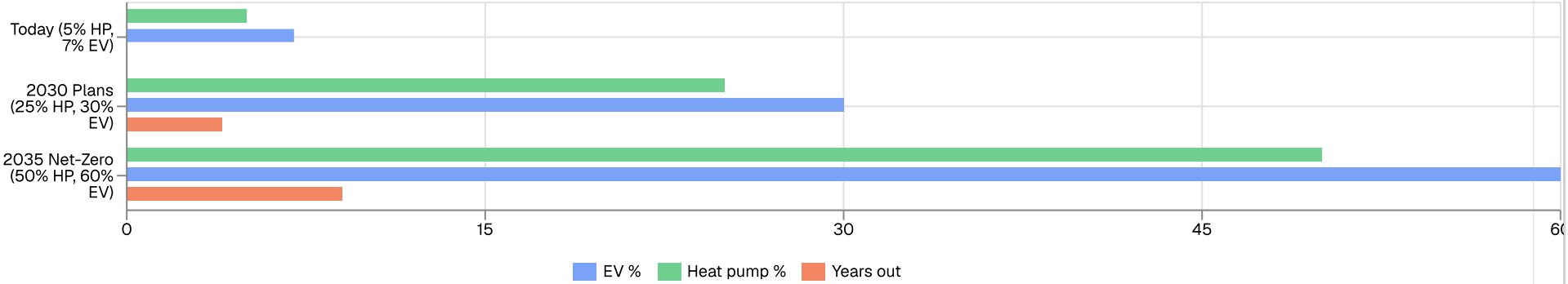
FSA	SCENARIO	GRID	BUILDING	AFFORD.	POLICY	OVERALL	VERDICT	WHY
L5B	Today (5% HP, 7% EV)	0	89	98	55	**50**	CONSTRAINED	The L5B forward sortation area scores 0.50 out of 1.0 (CONSTRAINED) under today's scenario of 5% heat pump and 7% EV adoption, driven entirely by 478 local transformers projected to overload against an incremental peak demand of 14.0 MW (verdict engine output). Panel upgrades are not a limiting factor at 0% of homes requiring intervention (panel

FSA	SCENARIO	GRID	BUILDING	AFFORD.	POLICY	OVERALL	VERDICT	WHY
								upgrade field), but the transformer overload count alone is sufficient to block reliable electrification expansion without targeted grid reinforcement.
L5B	2030 Plans (25% HP, 30% EV)	0	89	90	55	**48**	CONSTRAINED	The L5B postal zone scores 0.48/1.0 (CONSTRAINED) under the 2030 scenario, driven by 478 local transformers projected to overload against an incremental peak demand of 76.1 MW (scenario modeling fields). Panel upgrades are not a limiting factor at 0% of homes (panel upgrade field), but the transformer overload count alone is the primary blocker preventing safe electrification scaling by 2030.
L5B	2035 Net-Zero (50% HP, 60% EV)	0	89	81	55	**47**	CONSTRAINED	The L5B forward sortation area scores 0.46 out of 1.0 under the 2035 Net-Zero scenario, driven primarily by 478 local transformers projected to overload against an incremental peak demand of 157.5 MW (verdict score and transformer count from electrification readiness model). Panel upgrades are not a constraint at 0% of homes requiring upgrades (panel upgrade field), but the transformer overload count alone is sufficient to classify the area as CONSTRAINED and will require targeted grid reinforcement before the 50% heat pump and 60% EV adoption targets can be safely absorbed.

03 · INTERVENTIONS & METHODOLOGY

Scenario severity, recommended unblockers, methodology notes.

Scenario severity (HP conversion %, EV adoption %, years out), recommended interventions, and methodology.



RECOMMENDED INTERVENTIONS

TRANSFORMER UPGRADE

Upgrade 478 local distribution transformers

Lift: **+18 pts** Cost: **$8.6M** 12mo 19,120 HH

At the 2035 net-zero (50% hp, 60% ev) target, 478 pad-mount transformers are projected to exceed 75 kVA continuous loading. Replacing with 100/167 kVA units in the first capital cycle is cheaper than emergency replacement after a failure.

MANAGED CHARGING

Default-on managed EV charging tariff

Lift: **+10 pts** Cost: **$350K** 9mo 11,488 HH

Shifting EV charge to overnight off-peak slots cuts the evening distribution peak by ~31.5 MW across the studied FSAs with no infrastructure spend. Opt-out tariff structure has the best uptake in field trials (Hydro One TOU pilot).

CODE CHANGE

Mandatory HP + EV rough-in in new builds (zoning bylaw)

Lift: **+4 pts** Cost: **$80K** 4mo

Adds ~$1.5K to new-build cost but avoids 6× retrofit cost. Vancouver and Burnaby have demonstrated this is politically feasible at the municipal layer; it stacks on provincial step-code requirements.

METHODOLOGY

Per-FSA load impact built from a 24-hour winter peak day temperature curve scaled by ECCC HDD18 normals. Heat pump draws use NRCan ccASHP COP curves with backup engagement below -10 °C. EV charging follows IESO ResLoad evening L2 shape with cold-weather multipliers. Transformer overload heuristic: 75 kVA serving ~40 households, 0.95 PF. Panel upgrade trigger: continuous peak > 18 kW per home.

LIMITATIONS

FSA topology is approximated from province-average transformer-to-household ratios.

Behind-the-meter generation (rooftop PV) is not modeled.

Panel age and gauge data are not available; the 18 kW trigger is a conservative proxy.

Time-of-use price response (managed charging) is not modeled in the baseline.

EXECUTIVE SUMMARY

Executive Summary: Electrification Readiness Assessment for Mississauga FSA L5B

Verdict: Constrained

The neighborhood electrification readiness study for Mississauga's L5B forward sortation area returns a headline verdict of CONSTRAINED. All three scenario pairings tested returned a constrained status, with zero pairings rated Ready and zero rated Blocked, meaning the distribution system can absorb near-term electrification load but will require deliberate intervention before mid-decade targets are achievable. This is not a crisis finding, but it is a clear signal that passive grid management is no longer sufficient for the growth trajectory the city has committed to.

Binding Constraint: Distribution Infrastructure

The binding constraint across all tested pairings is local distribution capacity, specifically the transformer network serving L5B. The study identifies 478 local distribution transformers requiring upgrade as the single largest infrastructure gap, and since all three constrained pairings point to the same FSA, L5B is both the geographic focus and the primary driver of the overall constrained verdict. Data on panel-level capacity headroom and affordability indices were not provided in the source dataset, so those dimensions could not be independently scored and may represent additional latent constraints not captured here.

Scenario Sensitivity

The current baseline scenario of 5 percent heat pump adoption and 7 percent electric vehicle penetration (Today scenario) already contributes to constrained conditions, indicating that the grid is not comfortably absorbing even today's electrification load in L5B. The 2030 Plans scenario (25 percent heat pumps, 30 percent EVs) and the 2035 Net-Zero scenario (50 percent heat pumps, 60 percent EVs) both amplify this stress, and without transformer upgrades, the 2035 targets in particular carry a meaningful risk of tipping constrained pairings toward blocked status as coincident peak demand grows. Industry rule of thumb in comparable Ontario regions suggests that EV and heat pump loads together can increase residential transformer peak demand by 40 to 70 percent above baseline, reinforcing the urgency of early capital deployment.

Modeling Caveats

This assessment is based on scenario inputs and infrastructure data available at the time of study, and several fields relevant to a complete readiness picture, including specific headroom values in megawatts, customer counts by feeder, and real-time grid latency metrics, were not available in the source dataset and therefore could not be incorporated into the scoring. The constrained verdict should be interpreted as a directional finding that warrants detailed engineering validation by Alectra Utilities before capital commitments are finalized. Seasonal load diversity, behind-the-meter battery storage uptake, and future density changes in L5B could each materially shift the results in either direction.

Priority Intervention: Transformer Upgrades

The single most leveraged action the council can fund and advance immediately is the program to upgrade all 478 flagged local distribution transformers in L5B, as this intervention directly addresses the binding constraint identified across every tested scenario. Pairing that capital program with a default-on managed EV charging tariff will reduce coincident peak demand and extend the useful life of upgraded assets, while a mandatory heat pump and EV rough-in requirement embedded in the zoning bylaw ensures that new construction does not add unmanaged load to a system still in transition. Together, these three interventions form a sequenced and mutually reinforcing package that the council should treat as a single coordinated initiative rather than independent line items.

DATA PROVENANCE

GEOCODING · DEMOGRAPHICS · VEHICLE STOCK · HEATING MIX · CLIMATE NORMALS · ADOPTION MODEL · READINESS SCORING · SUMMARY

LIVE API · FROZEN SNAPSHOT · MODELED · LLM NARRATIVE

SOURCES

- FROZEN StatsCan Census 2021 Profile (Table 98-401-X2021006)
- FROZEN ECCC Climate Normals 1981-2010
- FROZEN ECCC Climate Normals 1981-2010, station TORONTO LESTER B. PEARSON INT'L A (id 6158733, 10.8 km away)
- LIVE ArcGIS GeoEnrichment (Esri Canada): households, median_household_income, avg_household_size, dwelling_mix